PROJECT DOCUMENTATION

06

File Reference and Code Modification Guide

Every tracked file, what it does, and the exact files to open for each kind of change — with the rules this build imposes on edits.

PREPARED FOR
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SITE
rishavprasad.in
Cloudflare Workers

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1.0

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01

How to use this guide

This document is the map from "I want to change something" to "these are the files, in this order." It describes the repository as handed over at commit `8c9ea09` on 31 October 2026.

- Paths are relative to the repository root. Commands are for Git Bash from that root.
- Numbered steps in a recipe are meant to be followed in order.
- Line numbers are a convenience for this commit; search the named symbol if a line has moved.
- No recipe requires editing generated output (`out/`) or touching Cloudflare resources unless it says so.

Two kinds of change

CHANGE	WHERE IT IS MADE	DEPLOY REQUIRED?	SPEED
Wording, projects, skills, résumé summary, awards	Admin dashboard at `/admin/`, then **Publish**	Yes, triggered by Publish	One workflow run
Layout, styling, new page, new endpoint, limits	Repository code	Yes, push to `main`	Typecheck, test, build, deploy
Secrets (admin password, Turnstile, GitHub token)	Cloudflare Worker settings	No	Takes effect immediately
Domain, DNS, HTTPS policy, DNSSEC	Cloudflare dashboard and registrar	No code change	Minutes to hours
Database schema	`migrations/` + a manual remote apply	Deploy code after the apply	Minutes

> **Content is not code.** The public pages read a build-time snapshot, `content/published.json`, which the admin dashboard refreshes through the API during a build. Editing markdown or JSON by hand does not update the live site; use the dashboard.

Reading conventions

- Paths such as `worker/index.ts` are click-level targets; a trailing `:12` is a line number at this commit.
- Blue boxes are notes about how the system works; red boxes mark a trap that has already caused a real failure.
- Figures are numbered by document; "Figure 6.1" is the first diagram in this document.
- Behaviour labels used in the audit — Working, Partially Working, Requires External Configuration — are defined in document 04.
- Commands assume Node.js 24 and the Wrangler pinned in the lockfile, run from the repository root.
- A change is finished when typecheck, tests and build pass locally and the deploy workflow is green.

02

Repository map

Excluding this documentation set, the repository tracks 86 files. Ten directories carry the whole system; the rest are configuration, assets and tests.

```
rishav-portfolio/
├── app/                    Next.js App Router — every page and the stylesheet
│   ├── (site)/             public pages (route group, no URL segment)
│   ├── admin/              admin shell at /admin/
│   ├── layout.tsx          root metadata, fonts, theme bootstrap, JSON-LD
│   ├── globals.css         design tokens, utilities, dark mode, print rules
│   ├── not-found.tsx       custom 404 (exported as out/404.html)
│   ├── robots.ts           /robots.txt
│   └── sitemap.ts          /sitemap.xml (route list is manual)
├── components/             React components, grouped by surface
│   ├── site/               nav, footer, theme toggle, contact form, resume print
│   ├── admin/              dashboard, per-tab editors, form fields
│   └── airshare/           browser AirShare app and its protocol constants
├── lib/                    shared configuration and typed content access
├── content/published.json  committed content snapshot used by the build
├── worker/                 Cloudflare Worker (Hono): API, auth, validation, AirShare
├── migrations/             D1 schema, applied manually and in order
├── public/                 static files served verbatim (images, résumé, headers)
├── scripts/                content fetch (prebuild) and password hashing
├── tests/                  Vitest suites and three production browser suites
├── airshare/               legacy Node LAN server, kept for reference and tests
├── documentation/          this documentation set and its sources
└── config                  next.config.ts, wrangler.jsonc, tsconfigs, package.json
```

DIRECTORY	WHAT LIVES THERE	SAFE TO EDIT?
`app/`	Routes and global styling	Yes, with the recipes below
`components/`	Client and server components	Yes
`lib/`	Site configuration, content types, icon map	Yes
`content/`	Build snapshot; overwritten by CI builds	Only as a local fallback
`worker/`	The only server-side code	Yes, then typecheck the Worker project
`migrations/`	Applied SQL history	Add files only; never edit applied ones
`public/`	Files served without the Worker	Yes; filenames become public URLs
`tests/`	Automated checks against local code and production	Yes; keep counts in the docs in sync
`airshare/`	Legacy Node server, not deployed	Rarely; it exists for the protocol test

03

How a request is served

There are only three serving paths, and which one answers a URL determines where a change belongs.

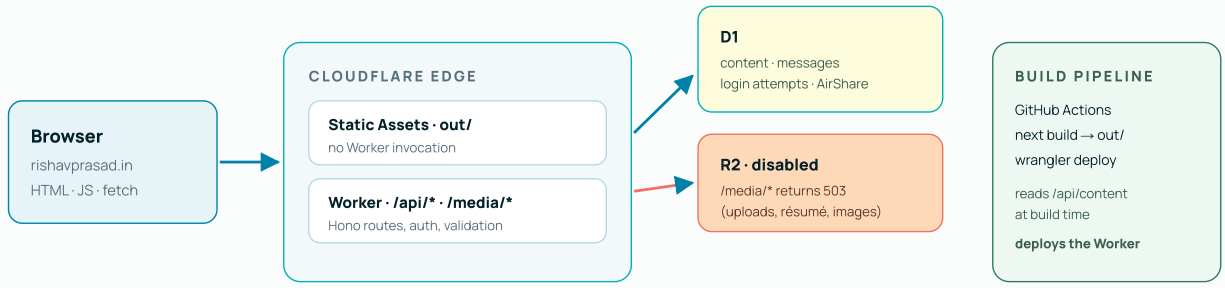


Figure 6.1 — Static files never reach the Worker; API and media paths always do. Content crosses from D1 into the static files only during a build.

URL PATTERN	ANSWERED BY	SOURCE FILES
`/`, `/about/`, `/projects/` …	Static Assets	`app/(site)/*/page.tsx` → `out/`
`/admin/`	Static shell + client calls	`app/admin/page.tsx`, `components/admin/*`
`/airshare/app/`	Static shell + client calls	`app/(site)/airshare/app/page.tsx`, `components/airshare/*`
`/api/health`, `/api/content`, `/api/contact`	Worker	`worker/index.ts`
`/api/airshare/*`	Worker	`worker/airshare.ts`
`/api/admin/*`	Worker, session cookie	`worker/index.ts`, `worker/auth.ts`
`/media/*`	Worker → R2	`worker/index.ts` (503 while R2 is off)
Any unknown path	Static Assets 404 page	`app/not-found.tsx` → `out/404.html`

04

Configuration and build files

FILE	WHAT IT CONTROLS	TOUCH IT TO…
`package.json`	Scripts and dependency ranges	Run `npm run build`, `test`, `typecheck`, `deploy`
`package-lock.json`	Exact versions; CI uses `npm ci`	Change dependencies with `npm install`, never by hand
`next.config.ts`	`output: "export"`, `trailingSlash`, unoptimized images	Change build output behaviour
`tsconfig.json`	App TypeScript, strict mode, `@/` path alias	Add path aliases or compiler options
`worker/tsconfig.json`	Worker TypeScript with Workers types	Worker-only compiler settings
`postcss.config.mjs`	Tailwind 4 PostCSS plugin	Rarely

FILE	WHAT IT CONTROLS	TOUCH IT TO…
`vitest.config.ts`	Unit test include patterns	Add a new test file location
`wrangler.jsonc`	Worker name, assets directory, routing, D1 binding, vars, custom domain	Change D1, domain, or enable R2
`public/_headers`	Security headers and Content-Security-Policy for static files	Allow a new external script or frame source
`.github/workflows/deploy.yml`	CI: install, typecheck, test, build, deploy	Change the pipeline or Node version
`.env.example`	Names of the values a local run needs	Document a new secret or variable
`.gitignore`	Excluded build output, local secrets, full database exports	Add a new local-only artifact

`.env.example` is a list of names, not a source of values. The site has no production `.env`. Worker variables are in `wrangler.jsonc`, secrets are stored by Cloudflare, and local development uses the git-ignored `.dev.vars`.

05

Pages, components and styles

Routes in `app/`

FILE	ROUTE	CONTENTS
`app/(site)/page.tsx`	`/`	Hero, portrait, focus areas, featured work
`app/(site)/about/page.tsx`	`/about/`	Bio, portrait, journey timeline, profile facts
`app/(site)/education/page.tsx`	`/education/`	Degree, board results, coursework notes
`app/(site)/skills/page.tsx`	`/skills/`	Skill groups and domains
`app/(site)/projects/page.tsx`	`/projects/`	Project cards with stack and status
`app/(site)/experience/page.tsx`	`/experience/`	Leadership roles and activities
`app/(site)/achievements/page.tsx`	`/achievements/`	Awards, achievements, certifications
`app/(site)/resume/page.tsx`	`/resume/`	Résumé summary, PDF link, print view
`app/(site)/airshare/page.tsx`	`/airshare/`	Explainer for the LAN transfer tool
`app/(site)/airshare/app/page.tsx`	`/airshare/app/`	Shell that mounts the client app
`app/(site)/contact/page.tsx`	`/contact/`	Contact form and direct email
`app/admin/page.tsx`	`/admin/`	Sign-in and dashboard shell
`app/layout.tsx`	all	Metadata, OG/Twitter tags, fonts, theme script, Person JSON-LD
`app/(site)/layout.tsx`	public	Skip link, navigation, footer
`app/not-found.tsx`	404	Custom not-found page
`app/globals.css`	all	Palette tokens, dark mode, buttons, layout utilities, print rules

FILE	ROUTE	CONTENTS
`app/robots.ts`, `app/sitemap.ts`	`/robots.txt`, `/sitemap.xml`	Crawler rules; sitemap route list is written by hand

Components

FILE	ROLE
`components/site/nav.tsx`	Sticky header, desktop links from `site.nav`, mobile sheet
`components/site/footer.tsx`	Contact email, social links, small print
`components/site/theme-toggle.tsx`	Light/dark switch; stores `theme` in localStorage
`components/site/contact-form.tsx`	Turnstile widget, validation, POST to `/api/contact`
`components/site/print-resume.tsx`	Print-only résumé layout used by `/resume/`
`components/site/timeline.tsx`	Timeline list primitives
`components/site/typing-roles.tsx`	Rotating role text on the home hero
`components/site/empty-state.tsx`	Shared empty-state block
`components/site/social-links.tsx`, `social-icon.tsx`	Social links and icons from content
`components/site/upcoming-page.tsx`	Placeholder page component; not imported by any current route
`components/admin/dashboard.tsx`	Tabs, loading/saving state, all admin API calls, Publish
`components/admin/content-editors.tsx`	One editor per tab: profile, education, skills, experience, projects, achievements
`components/admin/fields.tsx`	Inputs, text areas, list editors, repeatable entry lists
`components/airshare/airshare-app.tsx`	PIN room UI, WebRTC peer handling, chunked transfer, progress
`components/airshare/protocol.ts`	`CHUNK_BYTES` and `MAX_FILE_BYTES` shared by both sides
`components/static-link.tsx`	Plain `<a>` wrapper; the static export has no RSC payloads to prefetch

> **Keep `static-link`.** Next's client router expects RSC payload URLs that a static export does not emit, so `next/link` would 404 on navigation. Every internal link goes through the wrapper, which renders a normal anchor.

Design tokens in `app/globals.css`

TOKEN	LIGHT	USE
`--cerulean`	`#0081A7`	Primary accent, links beneath the surface, gradients
`--teal`	`#00AFB9`	Secondary accent, active underline, highlights
`--cream`	`#FDFCDC`	Page background
`--apricot`	`#FED9B7`	Warm tint blocks
`--coral`	`#F07167`	Calls to action, emphasis
`--ink`, `--muted`	`#0B2E38`, `#4C666E`	Body and secondary text
`--bg`, `--surface`, `--card`	cream, white, `#FFFDF2`	Surfaces; all redefined under `.dark`

Typography is Manrope for body and UI, Fraunces for display headings, both loaded by `next/font` in `app/layout.tsx`. There is no other palette; components use the tokens rather than raw hex values.

06

Worker, data, scripts and tests

FILE	ROLE
`worker/index.ts`	Hono app: security middleware, public API, contact, admin gate, media, error handlers
`worker/auth.ts`	PBKDF2 verification at 100,000 iterations, HMAC-signed session cookies
`worker/content-schema.ts`	Validation of the whole content document, per-field limits, safe link checks
`worker/airshare.ts`	Rooms, presence, signalling, per-IP and per-peer rate limits
`worker/env.ts`	Typed bindings: `DB`, optional `MEDIA`, and the five variables/secrets
`scripts/fetch-content.mjs`	Prebuild: fetch `/api/content`, write `content/published.json`; strict in CI
`scripts/hash-password.mjs`	Print an `ADMIN_PASSWORD_HASH` line without writing the password to disk
`migrations/0001_init.sql`	Content, contact messages, media, login attempts
`migrations/0002_airshare.sql`	Peers, signals, join attempts; TTL indexes
`tests/*.test.ts`	40 Vitest checks: auth, content validation, AirShare protocol
`tests/e2e.mjs`	31 public-route checks against a base URL
`tests/e2e-responsive.mjs`	84 width and page-error checks at seven viewports
`tests/e2e-airshare.mjs`	11 signalling checks using a transient room
`airshare/server.js`, `airshare/public/*`	Original Node LAN server; not deployed, used by `airshare/test/airshare.test.js`
`public/`	Portraits, OG image, favicon, résumé PDF, NPTEL certificate, `_headers`

The Worker's own constants sit at the top of each file and are the first place to look when behaviour looks surprising.

SETTING	VALUE AT HANDOVER	FILE
Session lifetime	12 hours	`worker/index.ts:10`
Request body cap	1 MB	`worker/index.ts:11`
Contact limit	5 per hour per IP hash	`worker/index.ts:12-13`
Login limit	8 failures per 15 minutes per IP hash	`worker/index.ts:14-15`
Content document cap	512 KB, 50 projects, 100 awards/certifications	`worker/content-schema.ts:33`
Room size and TTL	8 peers, 90 s presence, 10 min signals	`worker/airshare.ts:13-21`
Transfer size	64 KB chunks, 100 MB per file	`components/airshare/protocol.ts:1-2`

07

The content model and its two copies

One JSON document describes everything a visitor reads outside of layout: socials, education, skills, experience, projects, achievements, awards, certifications, résumé summary and profile facts. It exists twice.

COPY	LOCATION	WRITTEN BY	USED BY
Live	D1 `site_content.document` (row `id = 1`)	Admin **Save changes**	`GET /api/content`, admin editor
Build snapshot	`content/published.json`	`scripts/fetch-content.mjs`	Every public page at build time

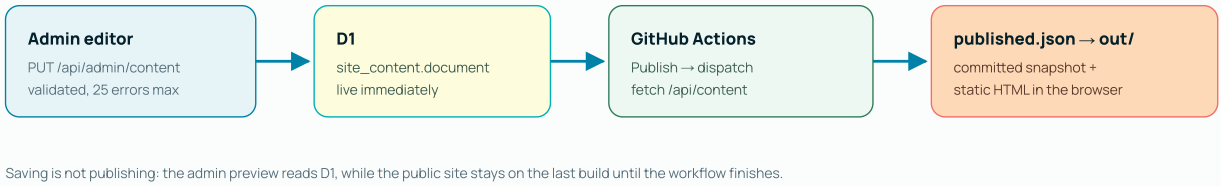


Figure 6.2 — Content path from editor to public HTML. The JSON file is an output of the build, not a source you maintain.

> **Do not hand-edit `content/published.json` and expect the site to change.** CI overwrites it from the API on every build. Use the dashboard. A manual edit is only a local fallback for offline builds, and it must be mirrored in the admin or it will disappear.

08

Modification recipes

Each recipe names the files to open and the follow-up that keeps the docs and tests truthful.

R1 · Change the palette or typography

1. Open `app/globals.css` and change the five palette tokens in `:root` (`--cerulean`, `--teal`, `--cream`, `--apricot`, `--coral`).
2. Adjust the derived surfaces in the same block: `--bg`, `--surface`, `--card`, `--ink`, `--muted`, `--line`, `--link`, `--nav`.
3. If dark mode should shift too, edit the `.dark` block underneath. Both blocks must stay complete; a missing token falls back to the light value.
4. Update the theme colour meta tags in `app/layout.tsx` (light `#FDFCDC`, dark `#04202A`) so mobile browser chrome matches.
5. Regenerate `public/images/og.jpg` if the social card shows the old palette.
6. Run `npm run build` and check one page in both themes at 375 px and 1440 px.

R2 · Add or rename a navigation link

1. Edit the `nav` array in `lib/site.ts` (line 22). Order matters; the header fits eight items at desktop width and collapses below 1280 px.
2. The `href` must match an exported route, with no trailing slash in the array (the site adds it in links and the router normalises it).
3. Add the same route to the `routes` list in `app/sitemap.ts` if it is a public page.
4. Rebuild and check the mobile menu, which renders the same array.

R3 · Add a new public page

1. Create `app/(site)/<slug>/page.tsx`. The route group does not appear in the URL.
2. Export `metadata` with a title and description; the root template appends the site name.
3. Read content through `lib/content.ts` rather than hardcoding strings, or use `site` from `lib/site.ts` for global text.
4. Add the route to `app/sitemap.ts` and, if it belongs in the header, to `site.nav`.
5. Add it to the route lists in `tests/e2e.mjs` and `tests/e2e-responsive.mjs` so the new page is covered.
6. Rebuild and confirm `out/<slug>/index.html` exists.

R4 · Change wording on the live site

1. Open `https://rishavprasad.in/admin/` and sign in.
2. Choose the matching tab: Profile, Education, Skills, Experience, Projects, Achievements, Resume.
3. Edit the fields and choose **Save changes**. The preview updates; the public site does not.
4. Choose **Publish**. A `content-published` workflow starts; wait for green, then hard-refresh.

> **When the dashboard has no field for it** — a new section, a layout change — it is a code change, not a content change. Follow R1–R3 or add the field to `components/admin/content-editors.tsx`, the type in `lib/content.ts`, and the validator in `worker/content-schema.ts` together.

R5 · Update the résumé PDF

1. Decide whether the new PDF is approved for publication; do not publish outdated or unapproved details.
2. **With R2 enabled:** Admin → Resume → choose the PDF → upload. The Worker stores it under `/media/` and sets the link.
3. **Without R2:** replace `public/resume/rishav-prasad-resume.pdf`, keep the same filename, and leave the content link unchanged.
4. Confirm the résumé page shows the new size and opens the file, and that the print view still fits one page.

R6 · Add a certificate or award

1. Put the PDF in `public/certifications/` using a lowercase hyphenated name.
2. Admin → Achievements → Certifications → *Add certification*.
3. Set `href` to `/certifications/<file>.pdf`. Relative paths are checked by the validator; external links must be `https`.
4. Save and publish. Open the link from `/achievements/` to confirm it resolves.

R7 · Enable R2 media

1. Enable R2 on the Cloudflare account and create the bucket `rishav-portfolio-media`.
2. Uncomment the `r2_buckets` block in `wrangler.jsonc` (line 29).
3. Deploy. `/media/missing` should now return 404 rather than 503.
4. Upload one image through Admin → Media and confirm it appears on the site when referenced.

R8 · Rotate the admin password

1. Run `node scripts/hash-password.mjs "<new long password>"` on its own and confirm the output starts with `pbkdf2_sha256$100000$`.
2. Run `npx wrangler secret put ADMIN_PASSWORD_HASH` and paste that line at the prompt. Never pipe the script into Wrangler: a refused password stores an empty secret.
3. Sign in with the new password. To end existing sessions, rotate `SESSION_SECRET` the same way.
4. No redeploy is needed; secrets are read at request time.

R9 · Change a rate limit or TTL

1. Find the constant at the top of `worker/index.ts` (contact, login, session) or `worker/airshare.ts` (rooms, signals).
2. Change the value and the test that asserts it in `tests/auth.test.ts`, `tests/content-schema.test.ts` or `tests/airshare-protocol.test.ts`.
3. Update `documentation/source/02-technical-specification.html` and the tables in document 04 so the numbers stay true.
4. Run `npm run typecheck && npm test`, then deploy.

R10 · Add an API endpoint

1. Add the route to `worker/index.ts`. Public routes go on `app`; anything touching content, messages or media goes on the `admin` router, which enforces the session cookie.
2. Validate input with the helpers in `worker/content-schema.ts`; never build SQL strings — use `.bind()`.
3. Return `{ "error": "…" }` with a status code, matching the existing routes.
4. Add the route to `API_DOCUMENTATION.md` and a check to `tests/e2e.mjs`.
5. If it needs a new binding, add it to `worker/env.ts` and `wrangler.jsonc`.

R11 · Change the schema

1. Add `migrations/0003_<name>.sql`. Never edit `0001` or `0002`; they are already applied.
2. Apply it locally with `npx wrangler d1 migrations apply rishav-portfolio --local`.
3. Apply it to production with `--remote` *before* deploying code that needs the new column.
4. Export a backup first if the migration rewrites data.

R12 · Allow an external script or frame

1. Add the origin to the `script-src` or `frame-src` list in `public/_headers`.
2. Keep the existing Cloudflare entries; Turnstile runs in an iframe from `challenges.cloudflare.com`.
3. Deploy and retest the contact form and the admin sign-in, which are the two pages that rely on third-party scripts.

09

Constraints every edit must respect

CONSTRAINT	WHY IT EXISTS	WHAT BREAKS IF IGNORED
`output: "export"`	The site is static files on Cloudflare, not a Node server	Server actions, route handlers and dynamic image optimisation do not exist at runtime
`trailingSlash: true`	Cloudflare serves folder indexes; the router is strict	Links without the slash 301 or 404 depending on the path
Plain anchors	No RSC payloads exist in the export	`next/link` navigation fails in the browser
Build-time content fetch	Pages read `published.json`, not the API	Editing D1 without publishing leaves the public site stale
PBKDF2 ≤ 100,000 iterations	Cloudflare rejects higher counts in Workers	Every sign-in returns “Invalid credentials”
Secrets only in Cloudflare	The repository is public	A committed secret must be rotated, not deleted
No localhost or preview URLs in content	Content is public and permanent	Broken links and an unprofessional public page
AirShare uses no STUN/TURN servers	It is deliberately LAN-only and PIN-gated	The app would silently depend on third-party infrastructure
Personal data stays out of git	Contact messages and full D1 exports contain a real name and email	Irreversible publication of someone else’s personal data

10

Verifying a change

Run these in order before pushing. The first three are also what CI runs, so a green local run predicts a green deploy.

COMMAND	CATCHES	TIME
`npm run typecheck`	Type errors in both the app and the Worker	Under a minute
`npm test`	Validation, auth and protocol regressions (40 checks)	Seconds
`npm run build`	Static export errors, missing routes, content fetch	About a minute
`npx wrangler dev --port 8787`	API behaviour with local D1 and `.dev.vars`	Manual
`E2E_BASE=https://rishavprasad.in node tests/e2e.mjs`	Production routes after a deploy (31 checks)	Under a minute

Stop the local server before rebuilding. Wrangler reads `out/` once at startup; rebuilding under a running server produces stale chunks and `ChunkLoadError`. Kill the whole Wrangler process tree, not one PID.

11

Common mistakes

MISTAKE	WHAT YOU SEE	FIX
Editing `published.json` by hand	Local change disappears after the next CI build	Make the change in Admin and publish
Piping the hash script into Wrangler	"Admin login is not configured yet" with the secret name present	Re-run `secret put` and paste the printed line
Generating a hash at 210,000 iterations	"Invalid credentials" for a correct password	Regenerate at 100,000
Using `next/link` or a relative image loader	Navigation 404s, images blank	Use `components/static-link` and files in `public/`
Forgetting the sitemap entry	New page is live but unlisted	Add the route to `app/sitemap.ts`
Editing an applied migration	Local and production schemas diverge	Add a new numbered migration
Adding a CDN script without a CSP entry	Script blocked on every page	Add the origin to `public/_headers`
Testing Turnstile on `127.0.0.1`	"Verification failed" locally only	Use `http://localhost:8787`
Uploading a file over 5 MB to media	Upload rejected	Compress the file or raise the Worker limit deliberately

12

Ownership and file sensitivity

PATH	STATUS	HANDLING
`.dev.vars`	Git-ignored	Local secrets only; never copy into the repo or a document
`backups/rishav-portfolio-*.sql`	Git-ignored	Full export with personal rows; keep off the repository and share only through private storage
`backups/schema-and-content-*.sql`	Committed	Sanitized: schema, migration history and the public content row
`content/published.json`	Committed	Snapshot; regenerated by the build
`out/`, `.next/`, `.wrangler/`	Git-ignored	Build and local runtime output; safe to delete and rebuild
`public/resume/`, `public/certifications/`	Committed	Public files; check for outdated or private details before committing
`documentation/`	Committed	This set plus its HTML sources and build script

The repository, Cloudflare account, GitHub account and domain registrar are all owned by Rishav Prasad. There is no shared, delegated or service account; CI uses a scoped API token stored as a repository secret.